Website Vulnerability Assessment Report

Security Analysis of Public Web Application

Target Website: ebieop.com

Assessment Type: Vulnerability Assessment

Prepared By: Aditya Sharma

March 2026

Executive Summary

This report presents the results of a vulnerability assessment conducted on ebieop.com.
The objective of the assessment was to identify potential security weaknesses in the publicly accessible web application.

The analysis was performed using passive security testing techniques and industry-standard tools.
Several configuration-related vulnerabilities were identified, primarily involving missing security headers and information disclosure.

These issues do not immediately compromise the system but could increase the attack surface if not addressed.
The report provides recommended remediation steps to improve the overall security posture of the website.

Scope of Assessment

The scope of this assessment included analysis of publicly accessible pages of the target website.

Testing was limited to passive security analysis and configuration review.
No intrusive testing, exploitation attempts, or denial-of-service activities were performed.

The objective was to identify potential vulnerabilities while maintaining ethical and non-disruptive testing practices.

The scope of this assessment included analysis of publicly accessible pages of the target website.
Testing was limited to passive security analysis and configuration review.
No intrusive testing, exploitation attempts, or denial-of-service activities were performed.

Tool Used

1. OWASP ZAP
2. NMap
3. Browser Developer Tool

These tools were used to perform passive scanning, identify security misconfigurations, analyze response headers, and detect potential vulnerabilities in the web application.

Network Scan Results

A network scan was performed to identify open ports and exposed services on the target system.

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV ebieop.com
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-12 07:47 EDT
Nmap scan report for ebieop.com (91.108.106.184)
Host is up (0.10s latency).
Other addresses for ebieop.com (not scanned): 2a02:4780:1f:6e95:92a5:da0b:19f8:2905 2a02:4780:20:2734:d420:9465:2728:65fd 2a02:4780:3c:bccc:877d:a8c6:212d:ae
dc 2a02:4780:1f:e39f:fd39:f2f7:259:ac01 2a02:4780:3c:7a70:7cf6:4db4:4472:58da
Not shown: 998 filtered tcp ports (no-response)
PORT    STATE SERVICE  VERSION
80/tcp  open  http     nginx (reverse proxy)
443/tcp open  ssl/http nginx (reverse proxy)
```

Figure 1: Nmap Scan Results showing open HTTP port

Findings Overview

ID	Vulnerability	Risk Level
1.	Missing X-Frame-Options Header	Medium
2.	Missing X-Content-Type-Options	Low
3.	Missing Strict-Transport-Security	Medium
4.	Server Information Disclosure	Low

FINDING 1

Title:

Missing X-Frame-Options Header

Content:

Risk Level: Medium

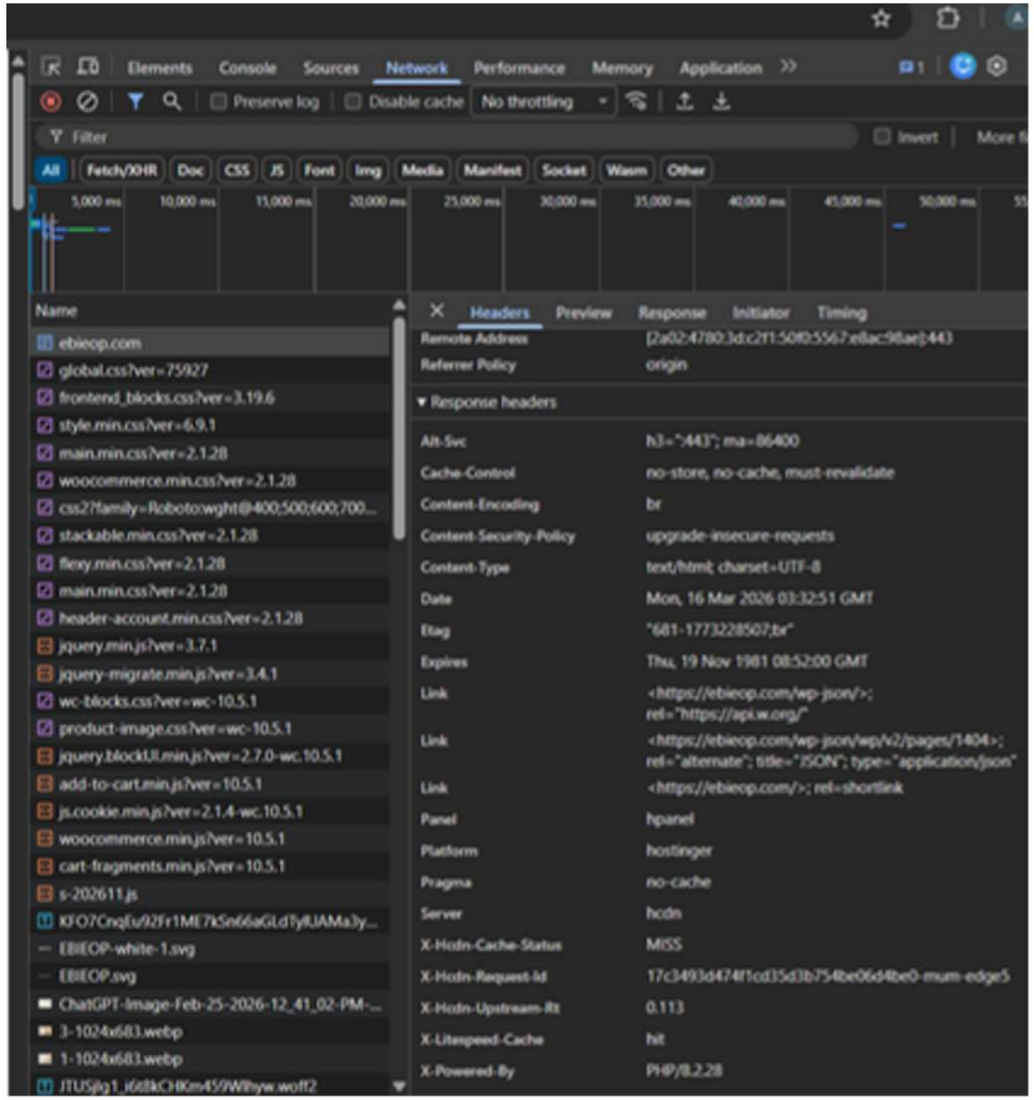
Description

The X-Frame-Options header helps protect websites against clickjacking attacks. Without this header, attackers could embed the website within malicious frames.

Remediation

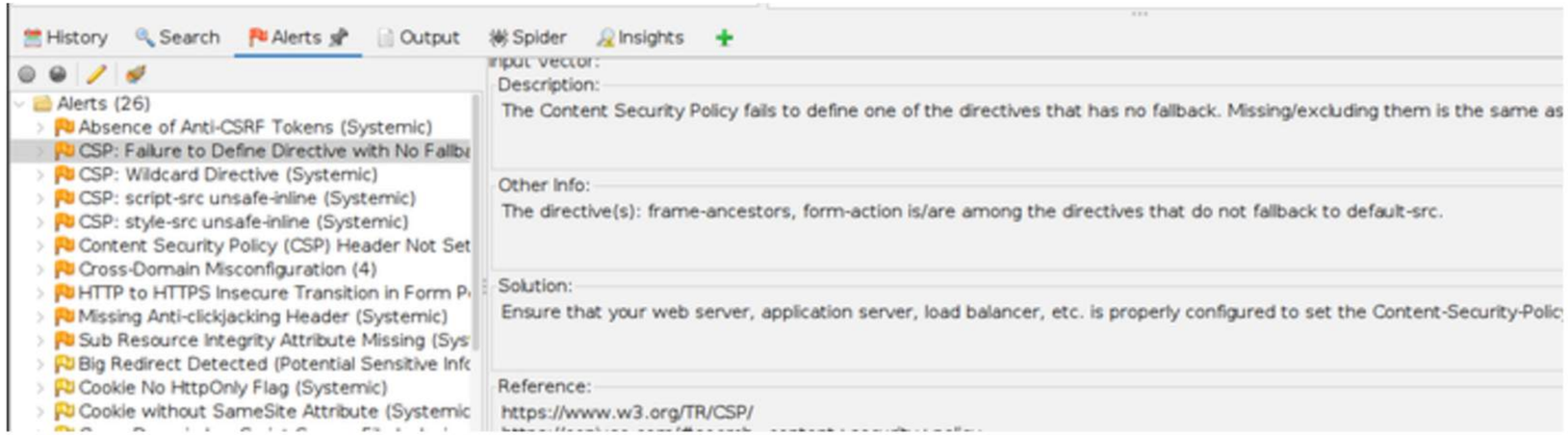
Add the following header in the web server configuration:

X-Frame-Options: SAMEORIGIN



Evidence

Finding 2



Title:

Missing X-Content-Type-Options Header

Risk:

Low

Description:

The X-Content-Type-Options header prevents browsers from MIME-type sniffing. Without it, browsers may interpret files incorrectly, potentially leading to security issues.

Fix:

X-Content-Type-Options: nosniff

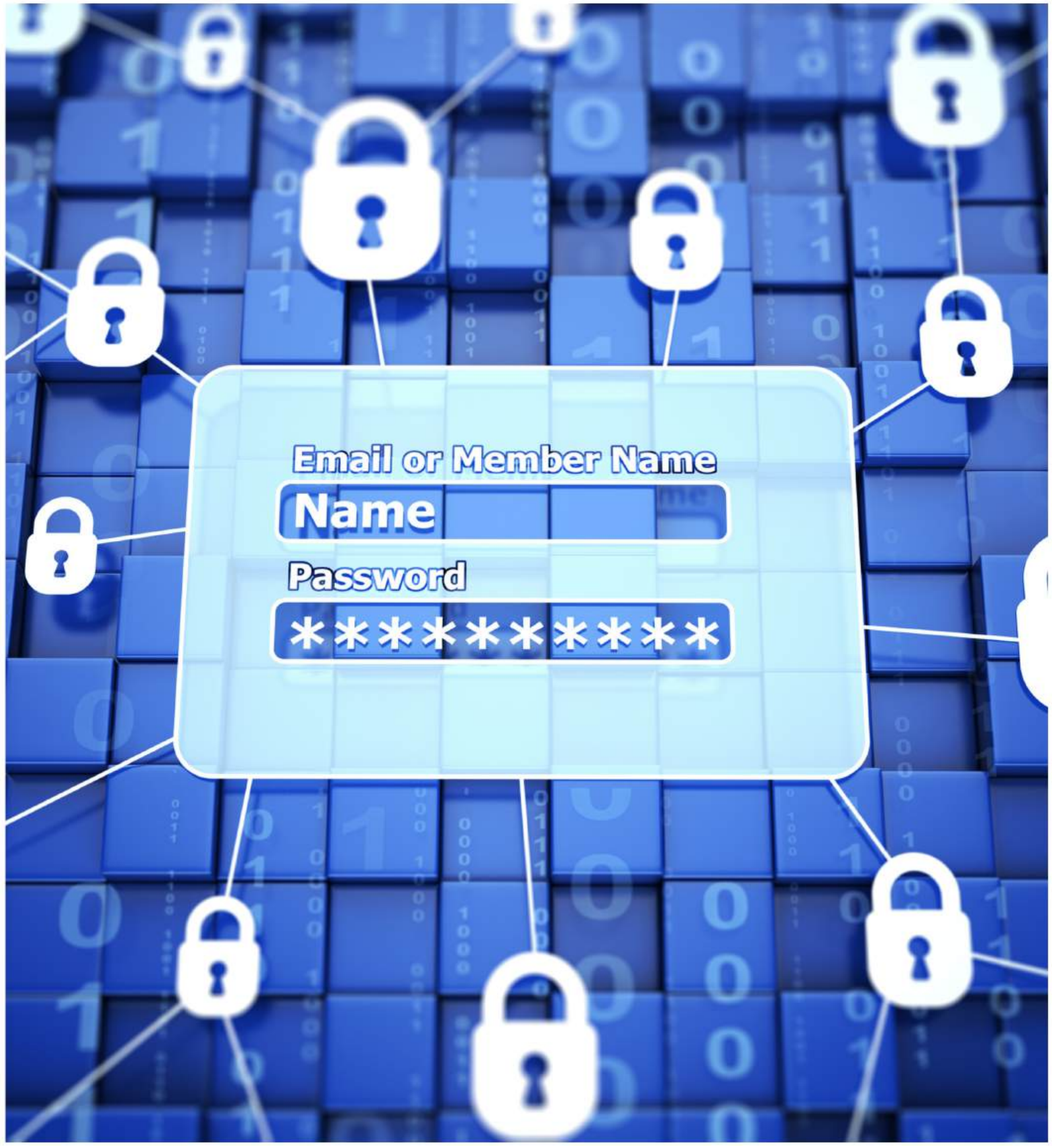
Finding 3

Title: Missing Strict-Transport-Security Header

Risk: Medium

Description: The Strict-Transport-Security (HSTS) header ensures browsers interact with the website only through secure HTTPS connections.

Fix: Strict-Transport-Security: max-age=31536000; includeSubDomains



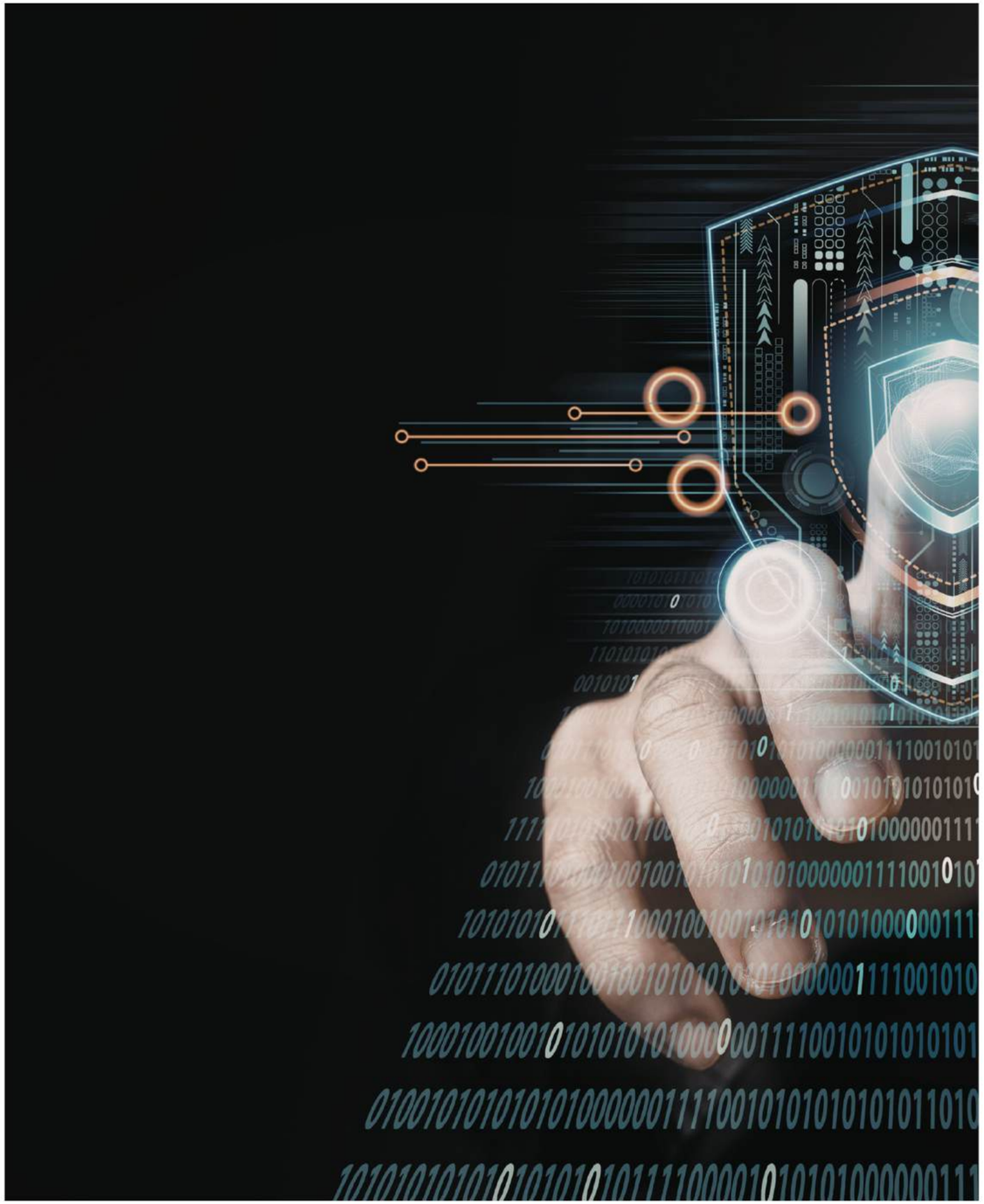
Finding 4

Title: Server Information Disclosure

Risk: Low

Description:
The response headers reveal server technology information such as PHP version
and hosting platform. Attackers may use this information to plan targeted attacks.

Fix: Disable unnecessary server information in headers.



CONCLUSION:

The vulnerability assessment identified several configuration-related security issues within the web application. While no critical vulnerabilities were detected,
the identified issues could potentially increase the attack surface of the system.

Implementing the recommended security headers and minimizing information disclosure
will significantly improve the security posture of the website.